PE Node KOR-PE-01 BGP Route Churn and OSPF Adjacency Loss SOP for Cisco CLI

Node: KOR-PE-01 | **Vendor:** Cisco | **Severity:** Warning | **Alarm Code:** ALARM_9548

Description

This Standard Operating Procedure (SOP) addresses the specific failure domain of Routing Protocol Flap (BGP route churn, OSPF adjacency loss, ISIS LSP flooding) on the PE node KOR-PE-01, running Cisco CLI at the Koramangala Datacenter. The PE node operates as a critical component of the Provider Edge (PE) device, facilitating seamless BGP routing, OSPF adjacency establishment, and ISIS LSP flooding. The BGP route churn and OSPF adjacency loss on this node can be attributed to the transient nature of BGP routing and OSPF adjacency establishment, which can be influenced by factors such as routing updates, network topology changes, or misconfigured routing policies. Additionally, ISIS LSP flooding can be triggered by network instability or misconfigured ISIS parameters. The Cisco CLI provides a robust platform for diagnosing and resolving these issues.

Pre-requisite Checks

- Check: Verify the Koramangala Datacenter is accessible and power is available to the PE node KOR-PE-01., Status: Completed

- Check: Confirm the redundant link status for the PE node KOR-PE-01., Status: Completed

Potential Root Causes

Cause	Probability
Misconfigured BGP routing policies	High
OSPF adjacency establishment issues due to network topology changes	Medium
ISIS LSP flooding triggered by network instability	Low

Diagnostic Commands & Expected Outputs

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Command: show ip bgp summary, Expected Output: BGP table version is 2, main routing table version is 2, Description: This command displays a summary of the BGP table, including the number of active paths and the number of updates received. It helps to identify any issues with BGP routing and adjacency establishment.

Command: show ip ospf neighbor, Expected Output: Neighbor ID Pri State Dead Time Address Interface, Description: This command displays a list of OSPF neighbors, including their priority, state, and dead time. It helps to identify any issues with OSPF adjacency establishment and network topology changes.

Command: show isis database brief, Expected Output: IS-IS Level-1 Database (Number, Sequence #, Metric, Next-Hop, Description: This command displays a list of ISIS LSPs, including their sequence number,
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metric, and next-hop. It helps to identify any issues with ISIS LSP
flooding and network instability.
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Resolution Steps

Step 1: Step: Verify and correct BGP routing policies, Command: show running-config | include bgp, Description: Verify the BGP routing policies and correct any misconfigurations to prevent BGP route churn and adjacency loss.

Step 2: Step: Verify and correct OSPF adjacency establishment, Command: show running-config | include ospf, Description: Verify the OSPF adjacency establishment and correct any misconfigurations to prevent adjacency loss due to network topology changes.

Step 3: Step: Verify and correct ISIS LSP flooding, Command: show running-config | include isis, Description: Verify the ISIS LSP flooding and correct any misconfigurations to prevent ISIS LSP flooding due to network instability.

Post-Check Verification

```
Command: show ip bgp summary, Expected Output: BGP table version is 2,
main routing table version is 2, Description: Verify BGP routing table
and adjacency establishment after resolving issues.

Command: show ip ospf neighbor, Expected Output: Neighbor ID Pri State
Dead Time Address Interface, Description: Verify OSPF adjacency
establishment and network topology after resolving issues.

Command: show isis database brief, Expected Output: IS-IS Level-1
Database (Number, Sequence #, Metric, Next-Hop, Description: Verify
ISIS LSP flooding and network stability after resolving issues.
```

Roll-back Procedures

- Step: Restore original BGP routing policies, Command: copy startup-config running-config, Description: Restore the original BGP routing policies to prevent further BGP route churn and adjacency loss.

- Step: Restore original OSPF adjacency establishment, Command: copy startup-config running-config, Description: Restore the original OSPF adjacency establishment to prevent further adjacency loss due to network topology changes.

- Step: Restore original ISIS LSP flooding, Command: copy startup-config running-config, Description: Restore the original ISIS LSP flooding to prevent further ISIS LSP flooding due to network instability.